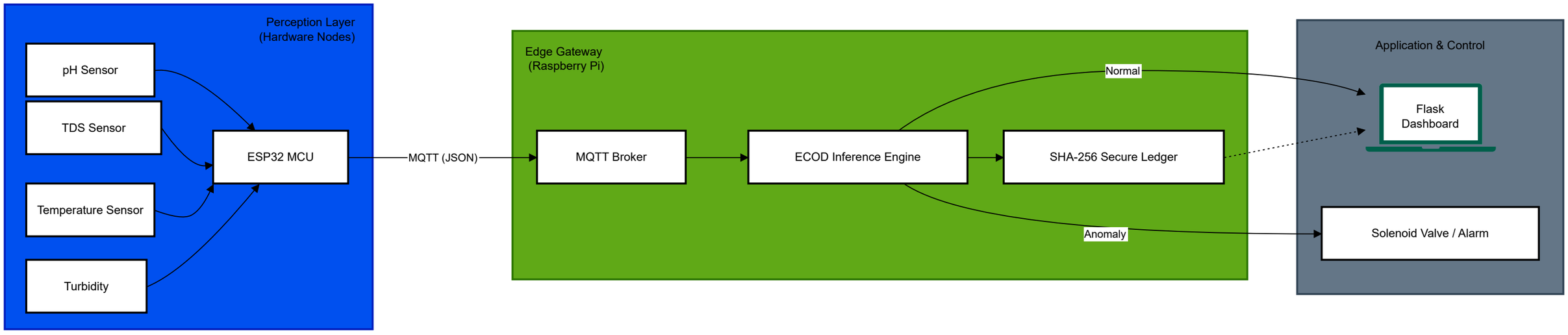System Architecture